

OVERVIEW

Data Scientist at Santander Corporate & Investment Banking (Global Headquarters, Madrid, Spain), leveraging statistical and data modeling to analyze equity derivatives and global markets data. Working on high-performance, scalable data pipelines to enhance quantitative models, trading operations, and strategic decisions for day-to-day trading desks.

EDUCATION

University of Pennsylvania (UPenn), Philadelphia *Aug 2025 – Aug 2027*
Master of Science in Engineering (MSE) in Data Science *CGPA: 4.00/4.00*

Narsee Monjee Institute of Management Studies (NMIMS), Mumbai *July 2019 – July 2023*
B.Tech (Honors) in Information Technology with Specialization in Machine Learning Models *CGPA: 3.60/4.00*

Relevant Coursework:

Machine Learning, Deep Learning, Natural Language Processing, Pattern & Anomaly detection, Data Structures & Algorithms, Database Management Systems, Computer Networks, Operating Systems, Math for ML - Linear Algebra, Probability & Statistics

PROFESSIONAL EXPERIENCE

Santander Corporate & Investment Banking *Data Scientist II (Pre/Post Trade Department)*
Warsaw, Poland – Remote *October 2024 – Present*

- Engineered scalable data pipelines via Apache Spark and Databricks on AWS EMR to process high-frequency market data for downstream quantitative modeling, ensuring ultra-low latency data delivery across global trading desks.
- Conducted time-series analysis on trade volumes and loss events, by utilizing Python and Statsmodels to identify anomalies and volatility trends, improving market risk assessments and strategic decision speed by 25%.
- Developed storage solutions (Scality S3, PostgreSQL) & performed statistical data profiling to ensure data integrity for equity derivatives, automating validation pipelines for millions of daily records, eliminating anomalies pre-deployment.
- Developed PowerBI and PowerApps monitoring real-time trade operations, incident management, and risk metrics.
- Implemented strict data governance and access controls to maintain regulatory compliance across business and quants.

Incedo Inc. *Software Engineer (Data & AI Department)*
Gurugram, India – on-site *July 2023 – October 2024*

- Developed "**BrainSpark**", an AI Agent facilitating automated testing script generation for software testing engineers
- Addressed manual HTML element detection by building a Selenium based XPath scraping system, which resulted in a significant reduction in the identification time by **50%** and streamlined QA workflows
- Generated AI-enabled QA workflows by using GPT-4o LLM, Retrieval-Augmented Generation(RAG), & VectorDB for semantic search, achieving **33% productivity gains** in the automation testing script generation

IBM (Co-ops) *Machine Learning Intern – (Data & AI)*
Remote *May 2022 – June 2022*

- Conducted secondary research, analyzed market trends, along with mapping customer journey & behavior
- Identified multiple touch points in the customer journey and recommended a 3-pronged strategy to optimize CX
- Built a multi-class classification model to analyze 10,000 customer verbatim messages and achieved an **82%** accuracy

SKILLS & TECHNOLOGIES

- **Languages:** Python, C++, SQL, Scala, C • **Libraries:** NumPy, Pandas, SciPy, Statsmodels, Scikit-learn, PySpark.
- **Technologies & Tools:** Matlab, Apache Spark, Hadoop, Kafka, AWS (EMR, EC2, Athena & Glue), PowerPlatform (PowerAutomate & PowerApps), PowerBI, SAS, Redhat openshift, Control M, LLMs(LangChain & RAG), Git(CI/CD)
- **Database & Management Tools:** PostgreSQL, Databricks, Scality S3, SharePoint, Liquibase database-versioning

LEADERSHIP & INTERESTS

- Core board member of Neurotech@Penn 2026; organized a global Ivy League neurotech hackathon in San Francisco.
- Research Lead, Claude Builder's Club at UPenn: Led sentiment analysis research on Penn students' LLM utilization.
- Collaborated with the Rotary Club of Udaipur to install foreign prosthetic hands for individuals missing upper limbs.
- Professional lawn tennis player: Underwent rigorous athletic training, representing the city in state-level tournaments.